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Cross-Validation

Video: Cross-Validation
6 min **Reading:** Summary
1h 30m

Module Assignment

Summary

Cross-validation is a commonly used technique for model evaluation and selection in machine learning. It helps to assess the performance of different models and select the one that generalizes well to unseen data. Ridge and Lasso regression are two regularization techniques used to address overfitting in linear regression models.

When applying cross-validation with Ridge and Lasso regression, the general procedure is as follows:

1. Split the dataset: Divide the available dataset into a training set and a validation set (also known as a holdout set). The training set is used to train the model, while the validation set is used to evaluate its performance.
2. Apply regularization: For Ridge regression, a regularization parameter (often denoted as α) is used to control the amount of regularization applied to the model. Similarly, Lasso regression employs the α parameter to determine the level of regularization. Different values of α can be tried during cross-validation to find the optimal value that minimizes the model's error.
3. Choose a cross-validation strategy: Select a suitable cross-validation strategy based on the size of your dataset and the characteristics of the problem. Commonly used techniques include k-fold cross-validation, stratified k-fold cross-validation, leave-one-out cross-validation, or time series cross-validation.
4. Perform cross-validation: Execute the chosen cross-validation strategy, which involves repeatedly fitting the Ridge or Lasso regression models on different subsets of the training data and evaluating their performance on the validation set. The performance metrics used can vary depending on the specific problem, but common ones include mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), or R-squared.
5. Select the best model: Calculate the average performance metric across all the cross-validation folds for each combination of α values. Choose the model with the best performance metric as the final model. The selected α value determines the level of regularization applied to the Ridge or Lasso regression model.
6. Evaluate the final model: Once the best model has been selected, it can be trained on the entire training set (including the validation set) using the chosen α value. Finally, evaluate the performance of the model on an independent test set or deploy it for predictions on new, unseen data.

By using cross-validation with Ridge and Lasso regression, you can find the optimal regularization parameter and select the model that balances the trade-off between bias and variance, leading to better generalization performance. Please see the attached readings that delves further into CV and its application in Ridge/Lasso Regression:

[Linear Lasso and Ridge Regression with R](#)

[Regularization in R Tutorial: Ridge, Lasso and Elastic Net](#)

[Supervised Statistical Learning Using Lasso Regression](#)

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